



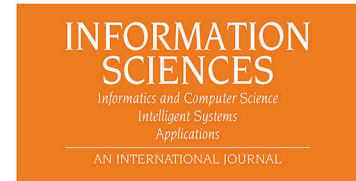
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Multi-Source Information Fusion Based Heterogeneous Network Embedding

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Abstract

Heterogeneous network embedding aims to learn a mapping between network data in original topological space and vectored data in low dimensional latent space, while encoding valuable information, such as structural and semantic information. The resulting vector representation has shown promising performance for extensive real-world applications, such as node classification and node clustering. However, most of existing methods merely focus on modeling network structural information, ignoring the rich multi-source information of different types of nodes. In this paper, we propose a novel Multi-source Information Fusion based Heterogeneous Network Embedding (MIFHNE) approach. We first capture the semantic information using the strategy of meta-graph based random walk. Subsequently, we jointly model the structural proximity, attribute information and label information in the framework of Nonnegative Matrix Factorization (NMF). Theoretical proofs and comprehensive experiments on two real-world heterogeneous network datasets demonstrate the feasibility and effectiveness of our approach.

Keywords: Heterogeneous Network Embedding, Multi-Source Information Fusion, Representation Learning, Joint Learning

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1. Introduction

A wide variety of Heterogeneous Information Networks (HINs) such as the Internet, social networks, bibliographic networks, protein-protein interaction networks, gene networks, and brain networks are ubiquitous in nature [1][2].

5 Particularly, online social network represented by Twitter, WeChat and Facebook has entered an era of billions of nodes. Therefore, the study of HINs is of great theoretical and practical significance[1].

Heterogeneous networks, as a kind of network data, are non-Euclidean data in the data science field. Unlike Euclidean data such as images where machine
10 learning methods can be conveniently applied, the high computational complexity and inapplicability of machine learning methods to network data have made network data mining and analysis tasks profoundly challenging [3]. Hence, network embedding technology came into being and has become an emerging research hotspot in recent years [3][4].

15 The target of network embedding is to learn a mapping between network data in original topological space and vectored data in low dimensional latent space. These vectored representations can preserve plentiful and valuable information, such as structural proximities between the nodes and multiple semantic information, which can be treated as feature vectors and applied in subsequent
20 extensive tasks such as node classification and node clustering via off-the-shelf machine learning technology [3][4].

Currently, researchers mainly study two types of network embedding algorithms: one is homogeneous network embedding, and the other is heterogeneous network embedding. The homogeneous network embedding algorithms are currently being studied more because the network is relatively simple compared
25 to heterogeneous networks. In the case of ignoring the node and edge type information, the homogeneous network embedding algorithm can be applied to heterogeneous networks. However, such direct application often results in poor performance. The heterogeneous network embedding algorithms are relatively
30 less studied, because designing efficient embedding algorithms for complex het-

erogeneous information networks is challenging [5].

Among existing heterogeneous network embedding algorithms, most algorithms, such as Metapath2vec [6] and Hebe [5], only model the structural information of the network. Moreover, very little efforts attempt to model multiple information in the network, such as the model proposed by H. Li et al. [7]. However, they have some defects, such as limited modeling information and high model complexity. It has been noticed that in actual heterogeneous networks, the node information can be very rich. In addition to semantic and structural information, valid information such as attributes and labels, also exists. All this information can be seen from multiple sources and converged into the network, and is referred to as, multi-source information in heterogeneous information network.

In this paper, we propose the incorporation of multi-source information to improve the performance of the heterogeneous network embedding algorithm and overcome the limitations of existing models. Our approach is inspired by the valuable insights. We take attribute information as an example. The different types of nodes in the heterogeneous networks with similar attributes are more likely to be associated with each other. For example, in online e-commerce network, buyers who prefer technology products and sellers that produce artificial intelligence (AI) chat robots are more likely to trade. In the field of scientific research, researchers in AI are more likely to publish their work in AI related journals and conferences primarily because the two types of nodes have similar attribute information. This indicates the importance of attribute information in heterogeneous network embedding to a certain extent. The significance of label information can also be deduced by the notion of analogy.

The contribution of this paper can be summarized in the following aspects:

- (1) We propose a novel and generic Multi-source Information Fusion based Heterogeneous Network Embedding (MIFHNE) approach in the framework of NMF. The framework has proven to be greatly efficient and has good scalability when additional information needs to be introduced.
- (2) Our approach jointly models rich semantics information, structural prox-

imity, attribute information and label information for learning informative node representations in heterogeneous information networks. To the best of our knowledge, this is the first heterogeneous network embedding method that considers multi-source information in the HINs.

(3) We prove the feasibility of the fusion of multi-source information for heterogeneous network embedding via rigorous theoretical analysis. We conduct comprehensive experiments with state-of-the-art homogeneous and heterogeneous network embedding baseline methods on two real-world heterogeneous network datasets. The empirical results demonstrate that our proposed MIFHNE approach achieves state-of-the-art performance on node classification and node clustering tasks.

The rest of the paper is organized as follows. Section 2 presents the related work. Section 3 describes some preliminary concepts and definitions formally. The proposed MIFHNE approach is presented in details in Section 4. In Section 5, comprehensive experiments are conducted on two publicly available datasets to validate our model. Finally, Section 6 concludes the literature and presents the future work.

2. Related work

In this section, we introduce the closely related work mainly from two branches: homogeneous network embedding methods and heterogeneous network embedding methods.

2.1. Homogeneous network embedding

From the perspective of modeling information, homogeneous network embedding algorithm can be divided into pure structural information-based model and multiple information-based model. The former can be mainly classified into four categories: random walk-based methods, optimization-based approaches, matrix factorization-based methods and deep learning-based methods. DeepWalk [8] and Node2Vec [9] are two representative methods of random walk-based

90 **approaches.** DeepWalk first generated sequences of nodes from a stream of truncated random walks on the graph. Then, node representation can be learnt, just like getting the representation of a word via word2vec [10]. Node2Vec improved DeepWalk via modifying the way of generating node sequences, which employed biased strategy for balancing the breadth-first sampling and depth-first sampling. 95 **Line [11] is the representative approach of optimization-based methods.** The model proposed the specific objective function to model the first-order proximity and second-order proximity of nodes. **The typical models based on matrix factorization are GraRep [12] and M-NMF [13].** GraRep presented a node embedding method by decomposing the global structural information matrix, which integrated various k-step relational information. M-NMF proposed a 100 modularized nonnegative matrix factorization model, which preserved both the microscope structure of first-order & second-order proximities and mesoscopic structure of community feature. **The fourth category is deep learning-based method. The typical works are DNGR [14] and ANE [15].** DNGR proposed a 105 deep graph representation method via applying stacked denoising autoencoders to the positive pointwise mutual information (PPMI) matrix, which is determined by the adjacency matrix. ANE designed a framework which mainly consists of two components, i.e., a structure preserving component and an adversarial learning component.

110 **The latter is multiple information-based model. The representative works are as following.** TADW [16] introduced text features matrix into matrix factorization framework. MMDW [17] incorporated the labeling information into vertex representations. TriDNR [18] learnt the graph representation from three parties, namely, node structure, node content, and node label. AANE [19] 115 proposed an accelerated attributed network embedding via enabling the joint learning process to be done in a distributed manner by decomposing the complex modeling and optimization into many sub-problems. ASNE [20] preserved the network topology and node attributes proximity through utilizing the multi-layer neural network.

120 Although the above methods have achieved impressive results in homoge-

neous network analysis tasks, they cannot be successfully applied to heterogeneous networks. If they are directly used to heterogeneous network, the performance of the algorithms will be greatly reduced. The main reason is that a large amount of semantic information composed of different types of nodes and relations is lost.

2.2. Heterogeneous network embedding

Heterogeneous network embedding algorithms can be mainly divided into four categories: optimized-based model, deep learning-based model, complex semantics-based model and multiple information-based model.

One of the primary work in the first category of methods is Latent Space Heterogeneous Model (LSHM) [21]. The model assumed that two interconnected nodes in the network tended to share similar representations regardless of their types. The node latent representation could be learnt by the function which combined the classification and regularization losses. The second category is deep learning-based model. The representative work is HNE [22], which jointly considered contents and topological structures in heterogeneous image-text networks. The model decomposed the feature learning process into multiple nonlinear layers of deep structure. The third class is complex semantics-based model, which contains several branches: hyperedge-based models, metapath-based models and extra models. Hebe [5] and DHNE [23] are the typical methods in the branch of hyperedge-based models. Hebe learnt multiple interactions in the heterogeneous information networks as a whole. The hyperedge was defined as a set of objects forming a consistent and complete semantic unit, which was taken as event. DHNE proposed to embed hypernetworks with indecomposable hyperedges. In the second class of metapath-based models, Metapath2vec [6], Metagraph2Vec [24] and HERec [25] are the typical methods. Metapath2vec proposed to preserve both structural and semantic correlations of heterogeneous graphs. However, modeling meta-paths in heterogeneous network was ineffective in capturing rich contexts and semantics between nodes. Therefore, Metagraph2Vec [24] and HERec [25] were proposed to improve the meta-path

based approaches. Moreover, there are some extra models. E.g. HENPC [26] extracted hierarchically organized, possibly overlapping and heterogeneous clusters, and exploited them for the task of multi-type clustering and classification in heterogeneous networks. InLPCH [27] constructed a triple heterogeneous network on the basis of the relationships between genes and lncRNAs and design a network propagation algorithm applied to the heterogeneous network for predicting disease-genes. HeteRank [2] integrated the multi-type relationships into similarity computation by utilizing all the meetings between objects for the task of similarity measure in heterogeneous information networks.

Note that most of the aforementioned approaches only capture network structural information and some semantics information while ignoring the rich attribute information and label information of different types of nodes. This may lead to the resulting node embedding vector, which contains limited information. Last, there are very little work, which incorporate multiple information to design heterogeneous network embedding model. The representative work was proposed by H. Li et al. [7]. However, the model may be greatly computationally time consuming if it stacks multilayer neural networks and the modeling of the information is limited.

3. Definitions

In this section, we formally summarize some notations and definitions used in this paper.

Definition 1: Network

Given a network $G = (V, E)$, V denotes the sets of nodes and E denotes the sets of edges between nodes. According to the directivity of edges, the network could be divided into undirected network and directed network. If the weight of edges is considered, the network can be further divided into weighted and unweighted network. By default, the object of this paper is undirected, unweighted network.

Definition 2: Homogeneous network, Heterogeneous network

180 Given $G = (V, E)$, $\Theta: V \rightarrow A$ and $\Psi: E \rightarrow R$ are type mapping functions for nodes and edges, respectively. $|A|$ and $|R|$ denote the sets of object and relation types, where $|A| + |R| > 2$ [6]. The network could be called a heterogeneous network. Otherwise, it is a homogeneous network.

Definition 3: First-order proximity

185 The first-order proximity is the weight $w_{u,v}$, between the edges of nodes u and v . It is the local pairwise proximity between two vertices. If there is no edge between nodes, then its first order proximity is 0 [11].

Definition 4: Second-order proximity, Higher-order proximity

The second-order proximity is the similarity of the first-order proximity of
190 node pairs [11]. Similar to the second-order proximity definition, the higher-order proximity is the similarity of the lower-order proximity.

Definition 5: Meta-path and Meta-graph of the network

Given a heterogeneous information network, $G = (V, E)$, a meta-path π is a sequence of node types $v_1, v_2, \dots, v_n$ and/or edge types $e_1, e_2, \dots, e_{n-1}$: $\pi = v_1$
195 $\xrightarrow{e_1} \dots v_i \xrightarrow{e_i} \dots \xrightarrow{e_{n-1}} v_n$ [6]. Meta-graph could be considered as a union of meta-path [24].

4. Method

In this section, we propose our novel and generic Multi-source Information Fusion based Heterogeneous Network Embedding (MIFHNE) approach in Non-negative Matrix Factorization (NMF) [28] framework, which could capture semantics information, structural proximity, attribute information and label information simultaneously. Fig. 1 shows the framework of our proposed MIFHNE approach. After learning the **informative node vector representations** which are marked as different colors in Fig. 1, **many off-the-shelf machine learning techniques such as Support Vector Machine (SVM) and K-means can be expediently**
205 **applied to heterogeneous network mining tasks efficiently.**

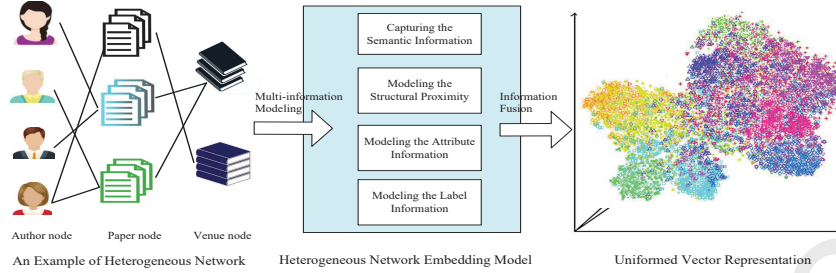


Fig. 1: The framework illustration of our proposed MIFHNE approach

4.1. Capturing Rich Semantics in Heterogeneous Network

Heterogeneous information network generally contains multiple semantic information. For capturing rich semantics in HINs, in this paper, we introduce a meta-graph based random walk strategy for heterogeneous network with multi-source information, which followed by the existing work (D. Zhang et al. [24], Y. Fang et al. [29], Y. R. Lin et al. [30]). Meta-graph is a union of meta-paths, which can capture more semantic relations between distant nodes [24]. The main difference is that each node in our strategy carries multiple source information. Formally, a meta-graph can be represented as $Metagraph = (V_n; E_n)$, where V_n represents different types of node sets, and E_n is the set of edges between V_n .

After extracting the multi-semantic information from heterogeneous network with multi-source information, we innovatively transform the original heterogeneous network into homogeneous network G_{homo} on the basis of the meta-graph strategy. Specifically, if node i is connected to node j through meta-graph, then the adjacency matrix of the network $Adj(i; j) = 1$. In this way, the homogeneous network can be constructed according to the semantics of meta-graph.

4.2. Modeling the Structural Proximity

For modeling the structural proximity, we propose to preserve the first-order proximity, second-order proximity and higher-order proximity for capturing structural information and attenuating sparse link problem concurrently

[31] [32] [13]. Formally, the first-order proximity is denoted by S^{first} , which equals the adjacency matrix $Adj(i, j)$. For node i , the first-order proximity with other nodes is denoted as $\Gamma_i = [S_{i1}^{first}, \dots, S_{in}^{first}]$. Therefore, the second-order proximity can be defined as

$$S_{ij}^{second} = \frac{\Gamma_i \Gamma_j}{\|\Gamma_i\| \|\Gamma_j\|}. \quad (1)$$

The higher-order proximities could be defined by using Katz Index [33] [34], which is formalized as

$$S_{ij}^{High} = (I - \mu Adj) \mu Adj, \quad (2)$$

where μ denotes a decay parameter and I represents the identity matrix. Based on the above definition, we could determine the final proximity matrix S as

$$S = S^{first} + \rho S^{second} + \eta S^{High}, \quad (3)$$

where ρ and η denote the balancing parameter. Then, we decompose the proximity matrix $S \in R^{n \times n}$ into a nonnegative basis matrix $M \in R^{n \times d}$ and a nonnegative node representation matrix $U \in R^{n \times d}$ in the framework of NMF, where d is the dimension of representation and the i -th row of $U(U_i)$ is the representation of node i . In order to approximate the similarity matrix S , we could define the following loss function as

$$L_{Struct} = \|S - MU^T\|_F^2 \quad s.t. \quad M \geq 0, U \geq 0. \quad (4)$$

225 4.3. Modeling the Label Information

For modeling the label information in the framework of NMF, we denote the node label matrix as $Z \in R^{1 \times n}$, where the i -th column of Z_i is the label of node i . Then, we approximate the node label matrix Z , by introducing a nonnegative basis matrix $E \in R^{1 \times d}$ and a nonnegative node representation matrix $U \in R^{n \times d}$. Consequently, the loss function could be defined as

$$L_{Label} = \|Z - EU^T\|_F^2 \quad s.t. \quad E \geq 0, U \geq 0. \quad (5)$$

4.4. Modeling the Attribute Information

Similarly, in order to model the attribute information in the framework of NMF, we could approximate the node attribute matrix B , by introducing a nonnegative basis matrix $N \in R^{m \times d}$ and a nonnegative node representation matrix $U \in R^{n \times d}$. The node attribute matrix can be denoted as $B \in R^{n \times m}$, where m is the dimension of representation and the i -th row of B_i is the attribute vector of node i . Therefore, we can define the loss function as

$$L_{Attri} = \|B^T - NU^T\|_F^2 \quad s.t. \quad N \geq 0, U \geq 0. \quad (6)$$

4.5. Joint Objective Function Fusing Multi-Source Information

For the sake of fusing multi-source information, we propose to incorporate the three models together so that the final model can capture structural proximity, attribute information and label information to jointly learn the better node representation matrix U . Accordingly, the final joint loss function could be defined as

$$L_{joint} = L_{Struct} + \alpha L_{Label} + \beta L_{Attri}, \quad (7)$$

where α and β are the balanced parameter. In this way, the matrix M, U, E, N can be solved by minimizing the final loss function L_{joint} . The mathematical form can be expressed as

$$\underset{M, U, E, N}{\operatorname{argmin}} L_{joint} \quad s.t. \quad M \geq 0, U \geq 0, E \geq 0, N \geq 0. \quad (8)$$

At this point, we have completed the design of heterogeneous network embedding algorithm by modeling multi-source information (semantic information, structural information, label information and attribute information) in heterogeneous network.

4.6. Optimization

Considering that the objective function is not convex, we decompose it into four sub-problems and adopt alternate optimization method, which guarantees each subproblem converges to the local minima [13][32].

4.6.1. Inferring Parameters

Update step for M : When fixing the other variables, updating M subproblem converts to a standard NMF optimization [28]. Therefore, the update rule for M are expressed as

$$M \leftarrow M \odot \frac{SU}{MU^TU}. \quad (9)$$

Update step for E : When fixing the other variables, updating E subproblem develops into a standard NMF optimization [28]. Therefore, the update rule for E are determined as

$$E \leftarrow E \odot \frac{ZU}{EU^TU}. \quad (10)$$

Update step for N : When fixing the other variables, updating N subproblem converts into a standard NMF optimization [28]. Therefore, the update rule for N are expressed as

$$N \leftarrow N \odot \frac{B^TU}{NU^TU}. \quad (11)$$

Update step for U : When fixing the other variables, updating U subproblem converts to a joint NMF problem [35]. Therefore, the update rule for U are determined as

$$U^T \leftarrow U^T \odot \left(\frac{M^TS + \alpha E^TZ + \beta N^TB^T}{M^TMU^T + \alpha E^TEU^T + \beta N^TN^TU^T} \right)^{\frac{1}{2}}. \quad (12)$$

4.6.2. Proof of convergence

Furthermore, for proving the validity of the update rule, we present a detailed convergence proof for the model via the following lemma. For update rules of M , E and N , they are the same type of standard NMF optimization problem. The specific proof could be followed by the previous work [28]. We will not repeat the relevant proof process and just focus on the proof of the update rule of node representation matrix U . The specific proof is as follows.

Lemma 1: The matrix A and X are non-negative, and $\tilde{X}$ is positive, then the following lemma could be

$$-tr(A^TX) \leq -tr(A^TP) - tr(A^T\tilde{X}), \quad (13)$$

where $P_{ij} = \tilde{X}_{ij} \ln X_{ij} / \tilde{X}_{ij}$. when $\tilde{X} = X$, the equation is satisfied.

Proof of Lemma 1:

$$\begin{aligned}
 \text{tr}(A^T X) &= -\text{vec}(A^T) \text{vec}(X) \\
 &\leq \text{vec}(A^T) \text{vec}(P) - \text{vec}(A^T) \text{vec}(\tilde{X}) \\
 &= -\text{tr}(A^T P) - \text{tr}(A^T \tilde{X})
 \end{aligned} \tag{14}$$

Lemma 2: The matrix A is non-negative, vector x is also non-negative, and $\tilde{x}$ is positive, then the following lemma could be

$$x^T A x \leq \frac{1}{2} c^T A \tilde{x} + \frac{1}{2} \tilde{x}^T A c, \tag{15}$$

where $c_i = x_i^2 / \tilde{x}_i$. When $\tilde{x} = x$, the equation is satisfied.

Proof of Lemma 2: Suppose $q_i = x_i / \tilde{x}_i$, then

$$\begin{aligned}
 &\frac{1}{2} c^T A \tilde{x} + \frac{1}{2} \tilde{x}^T A c - x^T A x \\
 &= \frac{1}{2} \tilde{x}^T Q^2 A \tilde{x} + \frac{1}{2} \tilde{x}^T A Q^2 \tilde{x} - x^T A x \\
 &= \frac{1}{2} \sum_{ij} A_{ij} \tilde{x}_i \tilde{x}_j (q_i - q_j)^2 \geq 0,
 \end{aligned} \tag{16}$$

when $\tilde{x} = x$, the equation is satisfied.

Lemma 3: The matrix A, D and X are non-negative, and $\tilde{X}$ is positive, then the following lemma could be

$$\text{tr}(D X^T A X) \leq \frac{1}{2} \text{tr}(D C^T A \tilde{X} + D \tilde{X}^T A C), \tag{17}$$

where $C_{ij} = X_{ij}^2 / \tilde{X}_{ij}$. When $\tilde{X} = X$, the equation is satisfied.

Proof of Lemma 3: According to Lemma 2, we can derive

$$\begin{aligned}
 \text{tr}(D X^T A X) &= \text{vec} X^T (D^T \otimes A) \text{vec}(X) \\
 &\leq \frac{1}{2} \text{vec}(C^T) (D^T \otimes A) \text{vec}(\tilde{X}) + \frac{1}{2} \tilde{X}^T (D^T \otimes A) \text{vec}(\tilde{C}) \\
 &= \frac{1}{2} \text{tr}(D C^T A \tilde{X} + D \tilde{X}^T A C),
 \end{aligned} \tag{18}$$

when $\tilde{X} = X$, the equation is satisfied.

Next, we deduce the updating rule of matrix U from Eq. (7) by using joint

nonnegative matrix factorization. The derivation steps are expressed as

$$\begin{aligned}
L_{joint} &= \|S - MU^T\|_F^2 + \alpha \|Z - EU^T\|_F^2 + \beta \|B^T - NU^T\|_F^2 \\
&= \text{tr}(S - MU^T) + \alpha \text{tr}(Z - EU^T) + \beta \text{tr}(B^T - NU^T) \\
&= \text{tr}(S^T S + \alpha Z^T Z + \beta B^T B) + \text{tr}(UM^T MU^T + \alpha UE^T EU^T + \beta UN^T NU^T) \\
&\quad - 2\text{tr}(S^T MU^T + \alpha Z^T EU^T + \beta BNU^T),
\end{aligned} \tag{19}$$

According to Lemma 3, we can get the following inequality,

$$\text{tr}(UM^T MU^T) \leq \frac{1}{2} \text{tr}(C^T M^T M\tilde{U}^T + \tilde{U} M^T MC) = \text{tr}(C^T M^T M\tilde{U}^T), \tag{20}$$

$$\text{tr}(UE^T EU^T) \leq \frac{1}{2} \text{tr}(C^T E^T E\tilde{U}^T + \tilde{U} E^T EC) = \text{tr}(C^T E^T E\tilde{U}^T), \tag{21}$$

$$\text{tr}(UN^T NU^T) \leq \frac{1}{2} \text{tr}(C^T N^T N\tilde{U}^T + \tilde{U} N^T NC) = \text{tr}(C^T N^T N\tilde{U}^T), \tag{22}$$

where $C_{ij} = (U^T)_{ij}^2 / \tilde{U}_{ij}^T$.

According to Lemma 1, we can also get the following inequality,

$$\text{tr}(S^T MU^T) \geq \text{tr}(S^T MP) + \text{tr}(S^T M\tilde{U}^T), \tag{23}$$

$$\text{tr}(Z^T EU^T) \geq \text{tr}(Z^T EP) + \text{tr}(Z^T E\tilde{U}^T), \tag{24}$$

$$\text{tr}(BNU^T) \geq \text{tr}(BNP) + \text{tr}(BN\tilde{U}^T), \tag{25}$$

250 where $P_{ij} = \tilde{U}_{ij}^T \text{In}(U_{ij}^T / \tilde{U}_{ij}^T)$. When $\tilde{U}^T = U^T$, the equation is satisfied.

According to formula (20-25), we can transform formula (19) into the following form,

$$\begin{aligned}
L_{joint} &\leq \text{tr}(S^T S + \alpha Z^T Z + \beta B^T B) + \text{tr}(C^T M^T M\tilde{U}^T) + \alpha \text{tr}(C^T E^T E\tilde{U}^T) \\
&\quad + \beta \text{tr}(C^T N^T N\tilde{U}^T) - 2\{\text{tr}(S^T MP) + \text{tr}(S^T M\tilde{U}^T)\} \\
&\quad - 2\alpha\{\text{tr}(Z^T EP) + \text{tr}(Z^T E\tilde{U}^T)\} - 2\beta\{\text{tr}(BNP) + \text{tr}(BN\tilde{U}^T)\} \\
&= \Gamma(U^T, \tilde{U}^T).
\end{aligned} \tag{26}$$

where $\Gamma(U^T, \tilde{U}^T)$ satisfies the condition as an auxiliary function of U^T (Please refer to the literature [39]). Then, Γ is nonincreasing under the update

$$(U^T)^{t+1} = \underset{U^T}{\operatorname{argmin}} \Gamma((U^T)^t, (U^T)^t). \quad (27)$$

According to the construction of auxiliary functions, we could get

$$\Gamma((U^T)^t, (U^T)^t) \geq \Gamma((U^T)^{t+1}, (U^T)^t) \geq \Gamma((U^T)^{t+1}, (U^T)^{t+1}). \quad (28)$$

Therefore, the loss function L_{joint} for $(U^T)^t = \Gamma((U^T)^t, (U^T)^t)$ is a monotonically decreasing function. According to the Karush-Kuhn-Tucker (KKT) conditions, we could get $\min_{U^T} \Gamma(U^T, \tilde{U}^T)$ as

$$\frac{\partial \Gamma}{\partial U_{ij}^T} = 2 \frac{U_{ij}^T}{\tilde{U}_{ij}^T} (M^T M \tilde{U}^T + \alpha E^T E \tilde{U}^T + \beta N^T N \tilde{U}^T)_{ij} - 2 \frac{\tilde{U}_{ij}^T}{U_{ij}^T} (M^T S + \alpha E^T Z + \beta N^T B^T)_{ij}. \quad (29)$$

Then, we can also get the update rule for U^T :

$$U^T \leftarrow U^T \odot \left(\frac{M^T S + \alpha E^T Z + \beta N^T B^T}{M^T M \tilde{U}^T + \alpha E^T E \tilde{U}^T + \beta N^T N \tilde{U}^T} \right)^{\frac{1}{2}}. \quad (30)$$

Therefore, Updating U with Eq.(12) will monotonically decrease the value of the objective in Eq.(7). Besides, theoretical analysis has shown that the update rule for M , E and N will also monotonically decrease the value of the objective in Eq.(7) [28]. Since the value of the objective in Eq.(7) is at least bounded
 255 by zero, we could finally conclude that the optimization method in our model converges.

Moreover, we further assess the convergence of our model from a purely experimental point of view, which is crucial to verify the correctness of the above theory and feasibility of the model. Without losing generality, we set
 260 the parameter α , β , μ , ρ , η as 10, 0.5, 0.1, 5, 1 for convergence analysis. The analysis results are reported in Figs. 2-3. From the results, we could distinctly observe that our proposed model could converge to a stable value on both data sets.

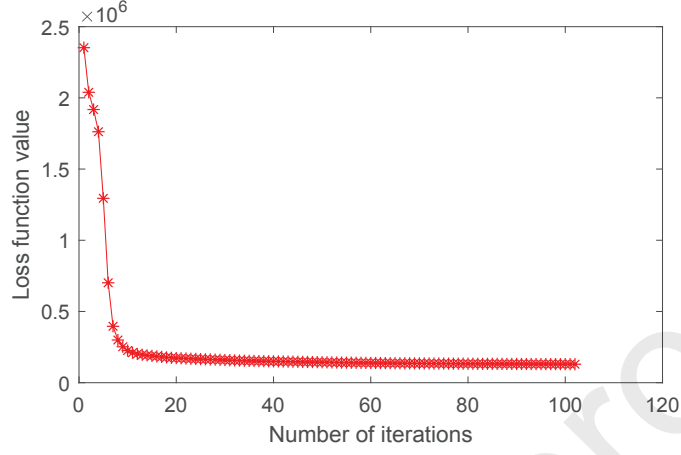


Fig. 2: Convergence analysis on the ACM datasets

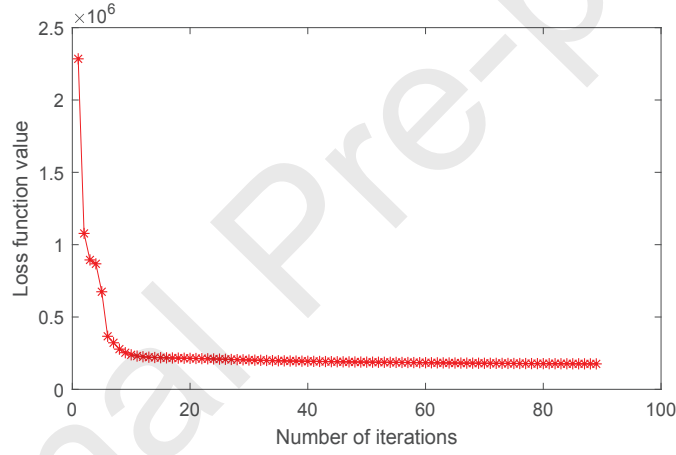


Fig. 3: Convergence analysis on the DBLP datasets

4.6.3. Proof of correctness

Excepting the proof of convergence, we also prove the correctness of the update rules of matrix U by introducing Lagrangian multiplier λ on matrix U^T . The Lagrangian function of Eq.(7) is expressed as

$$L = L_{joint} - tr(\lambda U). \quad (31)$$

Then, the derivative of $L(U^T)$ with respect to U^T is presented as

$$\frac{\partial L}{\partial U^T} = 2(M^T M U^T + \alpha E^T E U^T + \beta N^T N U^T) - 2(M^T S + \alpha E^T Z + \beta N^T B^T) - \lambda. \quad (32)$$

By setting the derivative to zero and using Karush-Kuhn-Tucker complementary condition [36], i.e., Eq.(32)=0, we get

$$\lambda_{ij}U_{ij}^T = [2(M^T MU^T + \alpha E^T EU^T + \beta N^T NU^T) - 2(M^T S + \alpha E^T Z + \beta N^T B^T)]_{ij}U_{ij}^T = 0. \quad (33)$$

265 We can find Eq.(12) satisfied the above Karush-Kuhn-Tucker complementary condition.

Last but no least, it is worth noting that the above four variables are randomly initialized and then the model is trained by iterative updating until the number of iterations reaches the predefined maximum number of iterations. Or, the loss should satisfy the following convergence condition. We further describe the pseudo code corresponding to our proposed method in Algorithm 1, which shows the core of our approach.

$$\left| \frac{L_{joint2} - L_{joint1}}{L_{joint1}} \right| < 1e - 3 \quad (34)$$

5. Experiments

270 (To evaluate the validity of the proposed MIFHNE approach, a supervised task and an unsupervised task on two real-world heterogeneous network datasets are conducted to examine the performance with the state-of-the-art methods, i.e., node classification and node clustering. The experiment code is compiled in MATLAB software. The experiment environment is on the Windows 10 Pro.64bit, CPU=Intel(R)Core(TM)i7-7500U CPU @2.70GHz and RAM=8GB.

5.1. Datasets

275 In order to ensure that the experimental heterogeneous network datasets with multi-source information, we employed the following two real-world network datasets, which are widely used in existing works [6][20]. The detailed statistics of the datasets are given in Table 1.

280 **DBLP Heterogeneous Network:** For DBLP dataset, we extract a subset of DBLP [37] bibliographic heterogeneous information network (DBLP-Citation-

Algorithm 1 Multi-source information fusion based heterogeneous network embedding.

Input: Structural proximity matrix, $S \in R^{n \times n}$;

Label information matrix, $Z \in R^{1 \times n}$;

Attribute information matrix, $B \in R^{n \times m}$;

Number of iterations, $Iter$;

Parameter $\alpha, \beta, \mu, \rho, \eta$

Output: Basis matrix, $M \in R^{n \times d}, E \in R^{1 \times d}, N \in R^{n \times d}$;

Embedding matrix, $U \in R^{n \times d}$;

1: Initialization: $M, E, U, N, Loss, MaxIter$;

2: **for** $i = 1$ to $MaxIter$ **do**

3: Update M based on Eq.(9);

4: Update E based on Eq.(10);

5: Update N based on Eq.(11);

6: Update U based on Eq.(12);

7: Calculating Loss based on Eq.(7);

8: **if** Loss satisfy convergence condition based on Eq.(34) **then**

9: save matrix U ;

10: break;

11: **end if**

12: **end for**

13: **return** U ;

network $V3^1$) being composed of authors (A) nodes, papers (P) nodes, venues (V) nodes and their relationships, which is similar to work [24]. According to the paper published in the different venues, we divided the papers into two research areas, namely, Data Mining and Artificial Intelligence. Specifically speaking, we choose two venues, namely, *SDM* and *COLT*. Then the relevant papers and authors establish corresponding relationships according to the venues. For

¹<https://www.aminer.cn/citation>

Table 1: Statistics of the heterogeneous network datasets.

Datasets	No. of nodes A	No. of nodes P	No. of nodes V	No. of edges
DBLP	1925	1511	2	5324
ACM	1942	1175	2	35307

capturing the rich semantic information, the random walk strategy based on the meta-graph composed of meta-path APAPA and meta-path APVPA is adopted [24]. We collect all the title content associated with one paper vertex, we likewise collect all the title content associated with papers which are written by one author, while the same setting is done to the venue vertex. Then the title text content of each node is taken as its corresponding attribute information. Finally, the vector representation of attribute information is obtained through the doc2vec model [38]. The dimension of attribute information is uniformly set to 200.

ACM Heterogeneous Network: For ACM dataset, we likewise extract a subset of heterogeneous information network (ACM-Citation-network V9²) which also contains authors (A) nodes, papers (P) nodes, venues (V) nodes and their relationships. Unlike the DBLP heterogeneous network, the venues we choose are *KDD* and *ICML*. The rest of the processing is in accordance with the DBLP heterogeneous network dataset.

5.2. Baseline Methods

In the selection of baseline methods, we scrutinize the representativeness of the method and the availability of its code, and exclude some methods in the field of multimedia such as Ognjen Arandjelovic, et al. [39]. Finally, we comprehensively compare the performance of our proposed MIFHNE with following nine state-of-the-art network embedding methods mainly from two aspects of homogeneous network and heterogeneous network. In the comparison of homogeneous network embedding algorithm, we constructed the homogeneous net-

²<https://www.aminer.cn/citation>

310 work from heterogeneous network based on processing of meta-graph random
walk strategy and network transformation. In the comparison of heterogeneous
network embedding, we employed the original heterogeneous network.

- Doc2Vec [38]: This is an advanced algorithm for mapping paragraphs to
vectors, which embeds any piece of text in a distributed vector using neural
315 network models.
- Node2Vec [9]: It learnt node embedding via employing the Skip-Gram
model [40] on the node sequences generated by potentially biased ran-
dom walk. The model controlled the random walk via two key hyper-
parameters p and q .
- 320 • Struc2vec [41]: The model leveraged a hierarchy to measure node simi-
larity at different scales, and constructed a multi-layer graph to encode
structural similarities and generate structural context for nodes.
- AANE [19]: This is an accelerated attributed network embedding method
being capable of capturing both network structure proximity and node
325 attributes proximity.
- Hebe [5]: The model learnt multiple interactions in the heterogeneous in-
formation networks to preserve more semantic information via introducing
the concept of hyperedge in the HIN [42]. Briefly, the core of the method
can be seen as a valuable extension of the existing model - Deepwalk [8]
330 and Line [11] from homogeneous networks to heterogeneous networks. Fol-
lowing the same process in the original paper, we also extract the events
information in the network in our experiment, which consists of four types
of objects: authors (A), terms (T), papers (P) and venues (V).
- HeteRank [2]: The model integrated the multi-type relationships in hetero-
335 geneous information network into node similarity computation by utilizing
all the meetings between objects. For obtaining the node embedding, we

first calculate the general relationship matrix and get the similarity measure. Then, we employ non-negative matrix factorization to determine node representations of different dimensions.

- 340 • Metapath2Vec [6]: The model formalized metapath-based random walks to construct the heterogeneous neighborhood of a node and then leveraged a heterogeneous Skip-Gram model to perform node embeddings. It is one of the state-of-the-art network embedding algorithms for HINs.
- 345 • Metagraph2Vec [24]: The model improved Metapath2Vec via augmenting multiple metapaths in HIN to capture more contexts and semantic relations between nodes.
- HERec [25]: It is the recent model for recommendation task. The core of the method is heterogeneous information network embedding. The model also adopted meta-path based random walk strategy to generate meaningful node sequences for conducting network embedding. The learned node embeddings were first transformed by a set of fusion functions, and subsequently integrated into an extended matrix factorization (MF) model. Afterwards, the extended MF model together with fusion functions were jointly optimized for the rating prediction task. In our paper, a variant of HERec with metapath2Vec is selected in order to facilitate experimental comparison. The final embedding is obtained via the linear fusion of the two meta-paths of APAPA and APVPA.

5.3. Node Classification

We first conduct a detailed experiment and analysis on the task of classification including homogeneous network embedding methods on node classification and heterogeneous network embedding methods on author classification. The experimental settings are similar to the previous work (Tang, Lei, et al. [43], Bryan Perozzi, et al. [8]). Specifically, we randomly sample a portion of the labeled nodes, and treat them as training data. The training ratios are set to 10%, 30%, 50%, 70%, 90% and 1%, 3%, 5%, 7%, 9% respectively. The rest of the

nodes are used as testing set. The ground truth of all the node label is given according to the filed of the venue [24]. We train the binary support vector machine (SVM) classifier using the fitsvm tools in Matlab. For the evaluation metric, we follow the literature [8], using Macro-F1 and Micro-F1. In our model, the attribute matrix of the node is normalized to 0-1. We set the parameter α , β , μ , ρ , η as 10, 0.5, 0.1, 5, 1 respectively. The dimension of each node representation is uniformly set to 128. We run each algorithms 10 times, and report the average performance. The comparison results are shown in Tables 2-5.

Table 2: Node classification performance comparison on DBLP dataset.

Algorithm	Doc2vec(2014)	Node2Vec(2016)	Struc2vec(2017)	AANE(2017)	MIFHNE
10%	Macro-F1	0.5443	0.6514	0.7448	0.7593
	Micro-F1	0.5653	0.6654	0.7478	0.7737
30%	Macro-F1	0.5897	0.6818	0.7695	0.8484
	Micro-F1	0.6044	0.6966	0.7709	0.8553
50%	Macro-F1	0.6116	0.7035	0.7828	0.8982
	Micro-F1	0.6237	0.7172	0.7842	0.9005
70%	Macro-F1	0.6298	0.7038	0.7838	0.9152
	Micro-F1	0.6408	0.7179	0.7853	0.9167
90%	Macro-F1	0.6340	0.7202	0.7892	0.9215
	Micro-F1	0.6430	0.7355	0.7922	0.9230

Table 3: Node classification performance comparison on ACM dataset.

Algorithm	Doc2vec(2014)	Node2Vec(2016)	Struc2vec(2017)	AANE(2017)	MIFHNE
10%	Macro-F1	0.4848	0.7361	0.6810	0.8792
	Micro-F1	0.9080	0.9382	0.9226	0.9644
30%	Macro-F1	0.5095	0.7956	0.7354	0.8920
	Micro-F1	0.9109	0.9481	0.9357	0.9682
50%	Macro-F1	0.5194	0.8066	0.7468	0.8937
	Micro-F1	0.9123	0.9489	0.9391	0.9693
70%	Macro-F1	0.5403	0.8103	0.7627	0.8959
	Micro-F1	0.9125	0.9515	0.9401	0.9696
90%	Macro-F1	0.5743	0.8204	0.7882	0.9013
	Micro-F1	0.9215	0.9522	0.9465	0.9712

As the result shows, our proposed MIFHNE outperforms most of state-of-the-art methods only except on ACM dataset in Table 5 when the training ratio is 1%. For example, in Tables 2-3, our algorithm raises the performance of Doc2vec by 36-45% in Macro-F1 and 26-34% in Micro-F1 on DBLP dataset. On ACM dataset, MIFHNE outperforms Node2vec over 7.80-49% in Macro-F1 and 4.72-24% in Micro-F1. We could also find that on two datasets, AANE

Table 4: Author classification performance comparison on DBLP dataset.

Algorithm		Metapath2Vec(2017)	Hebe(2017)	HeteRank(2018)	HERec(2018)	Metagraph2Vec(2018)	MIFHNE
1%	Macro-F1	0.4662	0.5179	0.6238	0.8412	0.8674	0.9011
	Micro-F1	0.6457	0.6240	0.6955	0.8750	0.8983	0.9109
3%	Macro-F1	0.6340	0.5222	0.6445	0.9351	0.9579	0.9612
	Micro-F1	0.7231	0.6278	0.7160	0.9403	0.9612	0.9642
5%	Macro-F1	0.6761	0.5295	0.6739	0.9375	0.9593	0.9745
	Micro-F1	0.7454	0.6341	0.7213	0.9431	0.9624	0.9765
7%	Macro-F1	0.7284	0.5348	0.6855	0.9398	0.9598	0.9808
	Micro-F1	0.7745	0.6342	0.7376	0.9454	0.9630	0.9822
9%	Macro-F1	0.7422	0.5676	0.6883	0.9437	0.9623	0.9881
	Micro-F1	0.7777	0.6400	0.7404	0.9488	0.9653	0.9889

Table 5: Author classification performance comparison on ACM dataset.

Algorithm		Metapath2Vec(2017)	Hebe(2017)	HeteRank(2018)	HERec(2018)	Metagraph2Vec(2018)	MIFHNE
1%	Macro-F1	0.4748	0.4732	0.4724	0.4775	0.5031	0.4787
	Micro-F1	0.8984	0.8829	0.8943	0.8992	0.9019	0.8867
3%	Macro-F1	0.4775	0.4733	0.4736	0.5460	0.5866	0.7605
	Micro-F1	0.8989	0.8943	0.8960	0.9082	0.9141	0.9334
5%	Macro-F1	0.4786	0.4737	0.5199	0.7345	0.7808	0.8574
	Micro-F1	0.8992	0.8983	0.9009	0.9348	0.9438	0.9551
7%	Macro-F1	0.4792	0.4776	0.5448	0.8033	0.8067	0.9421
	Micro-F1	0.8993	0.8986	0.9036	0.9465	0.9481	0.9788
9%	Macro-F1	0.4812	0.4831	0.6133	0.8447	0.8781	0.9577
	Micro-F1	0.8994	0.9002	0.9061	0.9550	0.9623	0.9837

outperforms the Node2Vec and Struc2vec. The main reason is due to the factor that compared with the latter (Node2Vec and Struc2vec) which only consider the network structural information, the former (AANE) further takes account of the attribute information.

385 Tables 4-5 show that the proposed method is still quite a bit competitive compared with Metapath2Vec, Hebe, HeteRank, HERec and Metagraph2Vec methods on two datasets. In brevity, the main difference between our model and baselines is that most of these methods only model the network structure information and ignore the use of multiple information in the heterogeneous
390 network.

5.4. Node Clustering

We also test how the latent representations learned by embedding methods can help the node clustering task in heterogeneous information networks. After getting the features of nodes, we use the classical K-means algorithm to
395 implement clustering. We run each methods 10 times and report the average

accuracy (ACC) index. Figs. 4-7 show the results of node clustering.

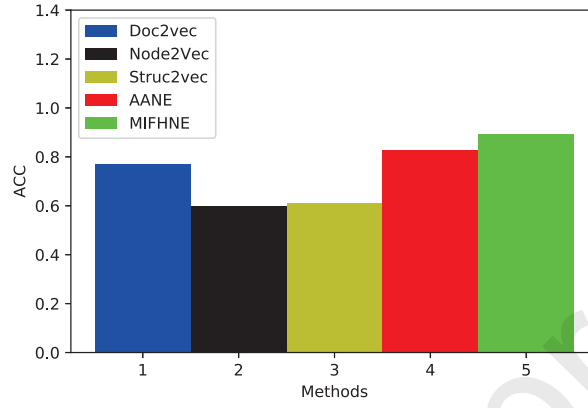


Fig. 4: Node clustering results on ACM dataset

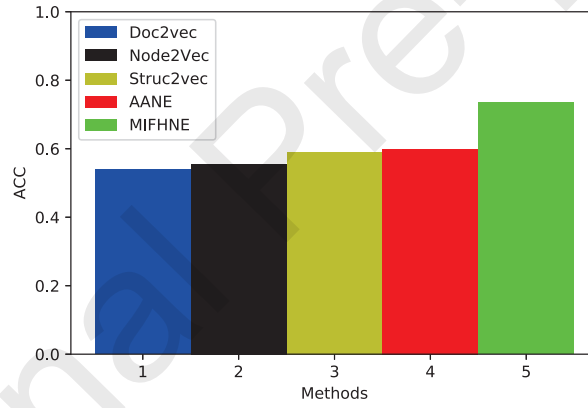


Fig. 5: Node clustering results on DBLP dataset

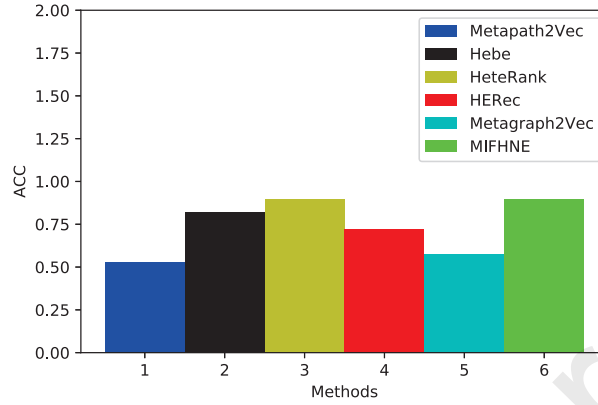


Fig. 6: Author clustering results on ACM dataset

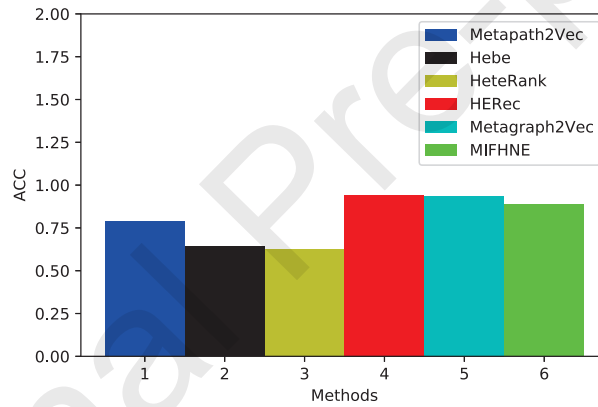


Fig. 7: Author clustering results on DBLP dataset

We can observe that our proposed method significantly surpasses all the comparative methods on node clustering task on ACM and DBLP datasets. On author clustering task, our algorithm outperforms the baseline methods on ACM dataset. On DBLP dataset, the proposed algorithm achieves competitive results. Overall, the above results validate the effectiveness of our proposed method on node clustering task.

5.5. Complexity Analysis

In this section, we analyze the complexity of our proposed MIFHNE model, which is dominated by the matrix computation in Formula (10), (11), (12), (13). The corresponding computation complexity is $\mathcal{O}(n^2d + d^2n)$, $\mathcal{O}(nd + d^2n)$, $\mathcal{O}(nmd + d^2n)$, $\mathcal{O}(n^2d + nd + nmd + d^2n + d^2 + d^2m)$, respectively. Since usually $d, m \leq n$, consequently, the overall computational complexity of MIFHNE is $\mathcal{O}(n^2d)$. It can be seen that the algorithm achieves a good compromise between complexity and performance.

5.6. Parameter Analysis

5.6.1. Impact of α and β

In this section, we study how the parameters α and β affect the performance of MIFHNE approach. We take the node classification task as an example. Both parameters are varied in the range of $\{0.5, 1, 5, 10\}$. We randomly select 10% of the dataset as training dataset and observe the change of score value of Macro-F1. The results are shown in Figs. 8-9.

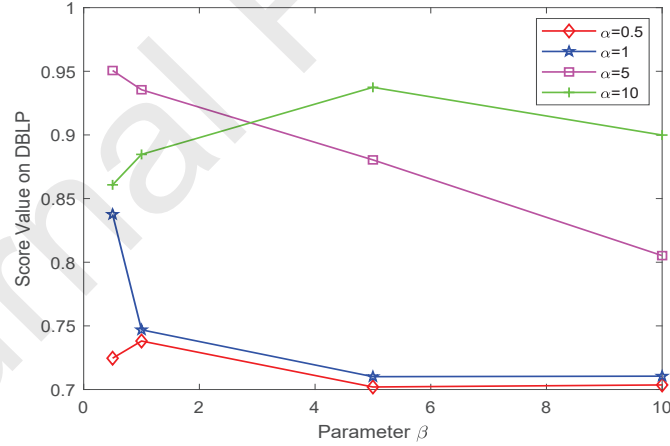


Fig. 8: Impact analysis of α and β on DBLP dataset

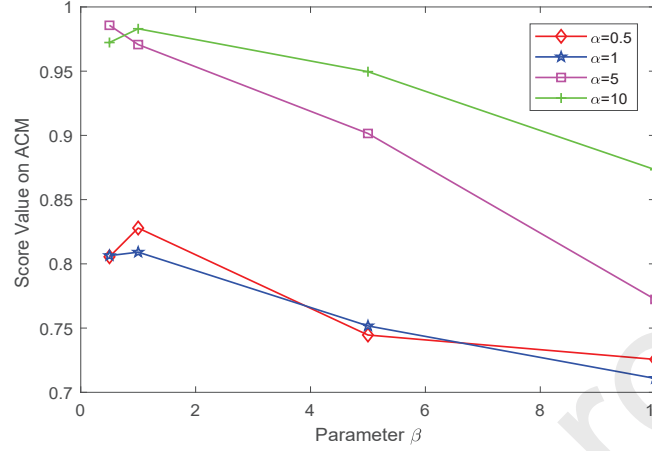


Fig. 9: Impact analysis of α and β on ACM dataset

We can observe that both parameters perform similar effect on the performance of our proposed model over the two datasets, except for $\alpha = 10$ on DBLP dataset. When β is greater than 1, the larger β is, the worse the node representation learned by the model is. And for α , the trend is opposite. Moreover, it is worth noting that when the parameter β is less than 1, the larger the parameter β is, the higher classification accuracy of the model could be. E.g. for two datasets, when $\alpha = 10$ and β ranges from 0.5 to 1, the score value shows an increasing trend.

Furthermore, we investigate the impact of $\alpha=0$ and/or $\beta=0$ on our model, which corresponded to the three situations: structure, structure+attribute, structure+label. We average the score value of Macro-F1 under the training ratio ranging from 10% to 90%. Fig. 10 shows the results of the analysis.

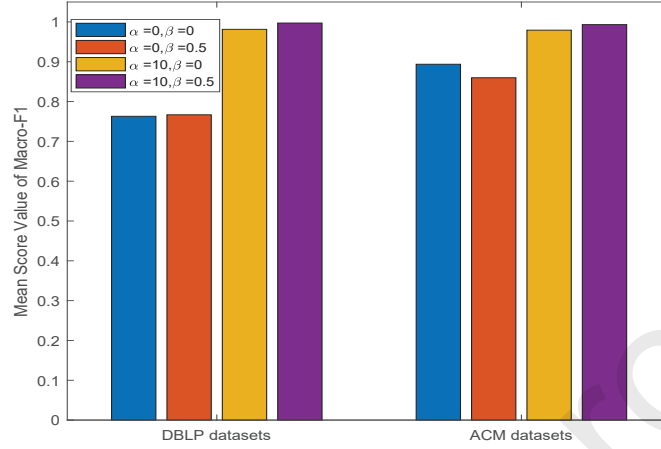


Fig. 10: Impact analysis of $\alpha=0$ and/or $\beta=0$ on two datasets

430 We can find that when $\alpha = 10$ and $\beta = 0.5$, the performance of our proposed MIFHNE approach is higher than that of other three situations. This further indicates the importance of multi-source information fusion on heterogeneous network embedding. Meanwhile, we can observe that for different datasets, there exist different complementary effects. Specifically, the score value becomes
 435 higher on DBLP dataset while the value gets lower on ACM dataset in two situations of structure and structure+attribute.

For node clustering task, we likewise analyze the similar complementary effect on ACM dataset. The results are shown in Fig. 11. We can find that the accuracy of clustering increases with the increase of the information involved in
 440 modeling.

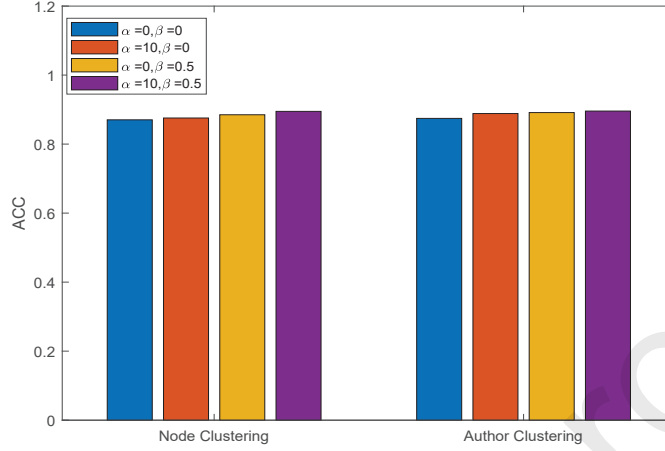


Fig. 11: Impact analysis of $\alpha=0$ and/or $\beta=0$ on ACM dataset

5.6.2. Impact of d

Finally, we study the impact of embedding dimension d , which varies within a certain range of $\{50, 90, 128, 170\}$ on all datasets. The training ratios for two datasets are set to 30% in the comparison with homogeneous network embedding baseline methods and 9% in the comparison with heterogeneous network embedding baseline methods respectively. The remaining experimental settings are the same as Section 5.3. The results are shown in following Figs. 12-15.

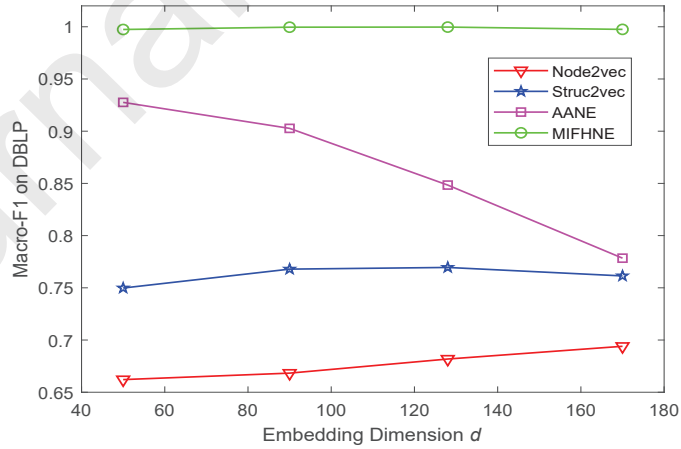
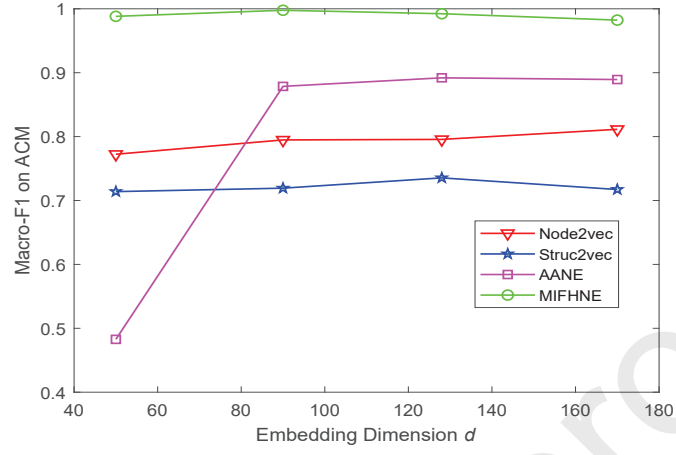
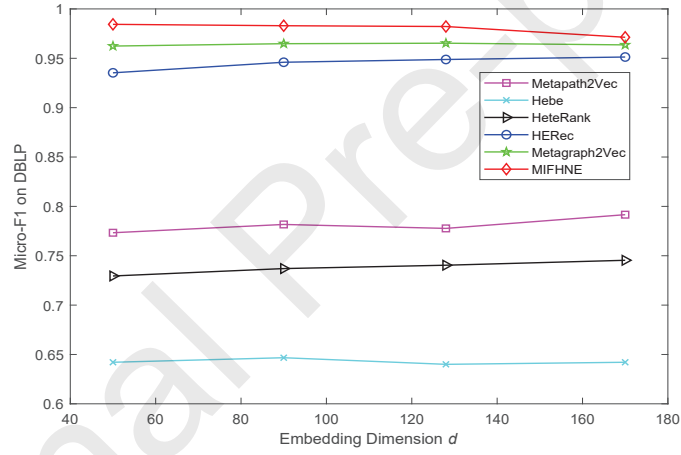


Fig. 12: Impact analysis of dimension d on DBLP dataset

Fig. 13: Impact analysis of dimension d on ACM datasetFig. 14: Impact analysis of dimension d on DBLP dataset

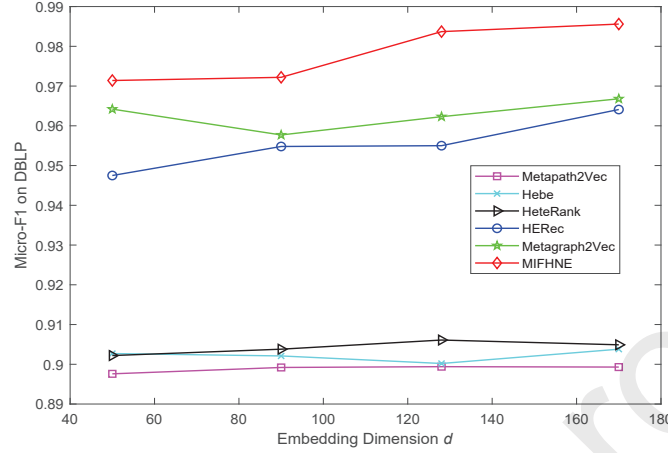


Fig. 15: Impact analysis of dimension d on ACM dataset

According to the above results, we can make three discoveries. In the first place, we could observe that MIFHNE approach consistently outperforms all baseline methods as d varies. In the next place, by increasing d , the performance of most methods keeps a steady and slightly increasing trend in addition to some algorithms such as AANE that are sensitive to dimensions on these two datasets. Lastly, the results indicate that the node vectors in low dimensional space obtained by our model have contained rich information.

Moreover, we also test the sensitivity of embedding dimension d on node clustering task. Without loss of generality, we study the impact on ACM dataset. Fig. 16 reports the analysis result. We can discovery that as the dimension d increases, the clustering performance curve has only slight fluctuation. In other words, the ACC value of our proposed method has been kept at a relatively high level.

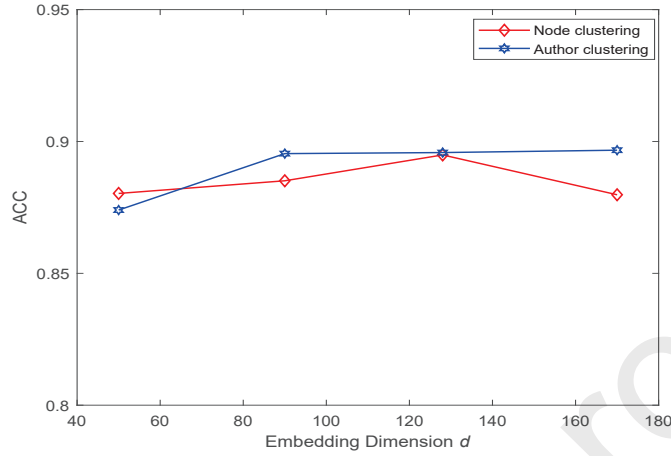


Fig. 16: Impact analysis of dimension d on ACM dataset

6. Conclusion

To learn informative node representations from heterogeneous network, it is crucial to intuitively fuse multiple information. In this literature, a novel heterogeneous network embedding method was proposed by fusing multi-source information in the HINs. The rich semantic information in HINs was firstly captured via the meta-graph based random walk strategy. Subsequently, the structural proximity, attribute information and label information were simultaneously modeled in the framework of NMF. This study not only proved the feasibility of the proposed MIFHNE model from the theoretical perspective, but also demonstrated the effectiveness of the model through comprehensive experiments. The proposed MIFHNE method was evaluated on two real-world heterogeneous network datasets and showed better performance by comparing with state-of-the-art homogeneous and heterogeneous network embedding baseline methods on the task of node classification and node clustering.

For future work, we will focus on the possibility of the attribute information of node being multimodal (e.g., image, text, and video). This will serve as a basis for designing an efficient embedding method for heterogeneous networks with multimodal attribute information.

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References

- [1] C. Shi, Y. Li, J. Zhang, Y. Sun, P. S. Yu, A survey of heterogeneous
information network analysis, *IEEE Trans. Knowl. Data Eng.* 29 (1) (2017)
17–37. doi:10.1109/TKDE.2016.2598561.
- 490 [2] M. Zhang, J. Wang, W. Wang, Heterank: A general similarity measure
in heterogeneous information networks by integrating multi-type relation-
ships, *Inf. Sci.* 453 (2018) 389–407. doi:10.1016/j.ins.2018.04.022.
- [3] P. Cui, X. Wang, J. Pei, W. Zhu, A survey on network embedding, *IEEE
Trans. Knowl. Data Eng.* 31 (5) (2019) 833–852. doi:10.1109/TKDE.2018.
495 2849727.
- [4] B. Li, D. Pi, Network representation learning: a systematic literature re-
view, *Neural Comput. Appl.* (2020).doi:10.1007/s00521-020-04908-5.
- [5] H. Gui, J. Liu, F. Tao, M. Jiang, B. Norick, L. M. Kaplan, J. Han, Em-
bedding learning with events in heterogeneous information networks, *IEEE
500 Trans. Knowl. Data Eng.* 29 (11) (2017) 2428–2441. doi:10.1109/TKDE.
2017.2733530.
- [6] Y. Dong, N. V. Chawla, A. Swami, metapath2vec: Scalable representa-
tion learning for heterogeneous networks, in: *Proceedings of the 23rd ACM
SIGKDD International Conference on Knowledge Discovery and Data Min-
505 ing*, 2017, pp. 135–144. doi:10.1145/3097983.3098036.

- [7] H. Li, H. Wang, Z. Yang, M. Odagaki, Variation autoencoder based network representation learning for classification, in: Proceedings of ACL 2017, Student Research Workshop, 2017, pp. 56–61.
- [8] B. Perozzi, R. Al-Rfou, S. Skiena, Deepwalk: online learning of social representations, in: The 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2014, pp. 701–710. doi:10.1145/2623330.2623732.
- [9] A. Grover, J. Leskovec, node2vec: Scalable feature learning for networks, in: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016, pp. 855–864. doi:10.1145/2939672.2939754.
- [10] T. Mikolov, K. Chen, G. Corrado, J. Dean, Efficient estimation of word representations in vector space, in: International Conference on Learning Representations, 2013.
- [11] J. Tang, M. Qu, M. Wang, M. Zhang, J. Yan, Q. Mei, Line: large-scale information network embedding, in: International World Wide Web Conferences, 2015, pp. 1067–1077. doi:10.1145/2736277.2741093.
- [12] S. Cao, W. Lu, Q. Xu, Grarep: Learning graph representations with global structural information, in: Proceedings of the 24th ACM International Conference on Information and Knowledge Management, 2015, pp. 891–900.
- [13] X. Wang, P. Cui, J. Wang, J. Pei, W. Zhu, S. Yang, Community preserving network embedding, in: Proceedings of the Thirty-First AAAI Conference on Artificial Intelligence, 2017, pp. 203–209.
- [14] S. Cao, W. Lu, Q. Xu, Deep neural networks for learning graph representations, in: Thirtieth AAAI Conference on Artificial Intelligence, 2016, pp. 1145–1152.

- [15] Q. Dai, Q. Li, J. Tang, D. Wang, Adversarial network embedding, in: Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence, 2018.
- [16] C. Yang, D. Zhao, D. Zhao, E. Y. Chang, E. Y. Chang, Network representation learning with rich text information, in: Proceedings of the Twenty-Fourth International Joint Conference on Artificial Intelligence, 2015, pp. 2111–2117.
- [17] C. Tu, W. Zhang, Z. Liu, M. Sun, Max-margin deepwalk: discriminative learning of network representation, in: International Joint Conference on Artificial Intelligence, 2016, pp. 3889–3895.
- [18] S. Pan, J. Wu, X. Zhu, C. Zhang, Y. Wang, Tri-party deep network representation, in: International Joint Conference on Artificial Intelligence, 2016, pp. 1895–1901.
- [19] X. Huang, J. Li, X. Hu, Accelerated attributed network embedding, in: SIAM International Conference on Data Mining, 2017, pp. 633–641. doi: 10.1137/1.9781611974973.71.
- [20] L. Liao, X. He, H. Zhang, T.-S. Chua, Attributed social network embedding, *IEEE Trans. Knowl. Data Eng.* 30 (12) (2018) 2257–2270. doi: 10.1109/TKDE.2018.2819980.
- [21] Y. Jacob, L. Denoyer, P. Gallinari, Learning latent representations of nodes for classifying in heterogeneous social networks, in: Proceedings of the 7th ACM international conference on Web search and data mining, 2014, pp. 373–382. doi:10.1145/2556195.2556225.
- [22] S. Chang, W. Han, J. Tang, G. Qi, C. C. Aggarwal, T. S. Huang, Heterogeneous network embedding via deep architectures, in: Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2015, pp. 119–128. doi:10.1145/2783258.2783296.

- [23] K. Tu, P. Cui, X. Wang, F. Wang, W. Zhu, Structural deep embedding for hyper-networks, in: Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence, 2018.
- [24] D. Zhang, J. Yin, X. Zhu, C. Zhang, Metagraph2vec: Complex semantic path augmented heterogeneous network embedding, in: Pacific-Asia Conference on Knowledge Discovery and Data Mining, 2018, pp. 196–208. doi:10.1007/978-3-319-93037-4_16.
- [25] C. Shi, B. Hu, W. X. Zhao, S. Y. Philip, Heterogeneous information network embedding for recommendation, IEEE Trans. Knowl. Data Eng. 31 (2) (2018) 357–370. doi:10.1109/TKDE.2018.2833443.
- [26] G. Pio, F. Serafino, D. Malerba, M. Ceci, Multi-type clustering and classification from heterogeneous networks, Inf. Sci. 425 (2018) 107–126. doi:10.1016/j.ins.2017.10.021.
- [27] X. Lei, Y. Zhang, Predicting disease-genes based on network information loss and protein complexes in heterogeneous network, Inf. Sci. 479 (2019) 386–400. doi:10.1016/j.ins.2018.12.008.
- [28] D. D. Lee, H. S. Seung, Algorithms for non-negative matrix factorization, in: Proceedings of the 13th International Conference on Neural Information Processing Systems, 2001, pp. 535–541.
- [29] Y. Fang, W. Lin, V. W. Zheng, M. Wu, C. C. Chang, X. L. Li, Semantic proximity search on graphs with metagraph-based learning, in: IEEE International Conference on Data Engineering, 2016, pp. 277–288. doi:10.1109/ICDE.2016.7498247.
- [30] Y. R. Lin, J. Sun, H. Sundaram, A. Kelliher, P. Castro, R. Konuru, Community discovery via metagraph factorization, ACM Trans. Knowl. Discov. Data 5 (3) (2011) 1–44. doi:10.1145/1993077.1993081.

- [31] H. Cai, V. W. Zheng, C. C. Chang, A comprehensive survey of graph embedding: Problems, techniques and applications, *IEEE Trans. Knowl. Data Eng.* 30 (9) (2018) 1616–1637. doi:10.1109/TKDE.2018.2807452.
- [32] J. H. Li, C. D. Wang, L. Huang, D. Huang, J. H. Lai, P. Chen, Attributed
590 network embedding with micro-meso structure, in: *International Conference on Database Systems for Advanced Applications*, 2018, pp. 20–36. doi:10.1007/978-3-319-91452-7_2.
- [33] L. Katz, A new status index derived from sociometric analysis, *Psychometrika* 18 (1) (1953) 39–43. doi:10.1007/BF02289026.
- [34] P. Goyal, E. Ferrara, Graph embedding techniques, applications, and performance: A survey, *Knowledge-Based Syst.* 151 (2018) 78–94. doi:10.1016/j.knosys.2018.03.022.
- [35] Z. Akata, C. Thureau, C. Bauckhage, Non-negative matrix factorization in multimodality data for segmentation and label prediction, in: *16th Computer Vision Winter Workshop*, 2011.
600
- [36] S. Boyd, L. Vandenberghe, *Convex optimization*, Cambridge university press, 2004.
- [37] J. Tang, J. Zhang, L. Yao, J. Li, L. Zhang, Z. Su, Arnetminer: extraction and mining of academic social networks, in: *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2008, pp. 990–998.
605 doi:10.1145/1401890.1402008.
- [38] Q. V. Le, T. Mikolov, Distributed representations of sentences and documents, in: *Proceedings of the 31th International Conference on Machine Learning*, 2014, pp. 1188–1196.
- [39] O. Arandjelovic, Weighted linear fusion of multimodal data: A reasonable baseline?, in: *Proceedings of the 2016 ACM Conference on Multimedia Conference*, 2016, pp. 851–857. doi:10.1145/2964284.2964304.
610

- [40] T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, J. Dean, Distributed representations of words and phrases and their compositionality, in: 27th
615 Annual Conference on Neural Information Processing Systems, 2013, pp. 3111–3119.
- [41] L. F. R. Ribeiro, P. H. P. Saverese, D. R. Figueiredo, *struc2vec*: Learning node representations from structural identity, in: Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data
620 Mining, 2017, pp. 385–394. doi:10.1145/3097983.3098061.
- [42] J. Silva, R. Willett, Hypergraph-based anomaly detection of high-dimensional co-occurrences, IEEE Trans. Pattern Anal. Mach. Intell. 31 (3) (2009) 563–569. doi:10.1109/TPAMI.2008.232.
- [43] L. Tang, H. Liu, Relational learning via latent social dimensions, in: ACM
625 SIGKDD International Conference on Knowledge Discovery and Data Mining, 2009, pp. 817–826. doi:10.1145/1557019.1557109.